All gears in the shafts 1,2& 3,4 & 5,6 & 7,8,9 rotate. However the main output shaft does not rotate as the dog clutches are not in engagement with any of the gears. Hence rotational power is not transmitted.

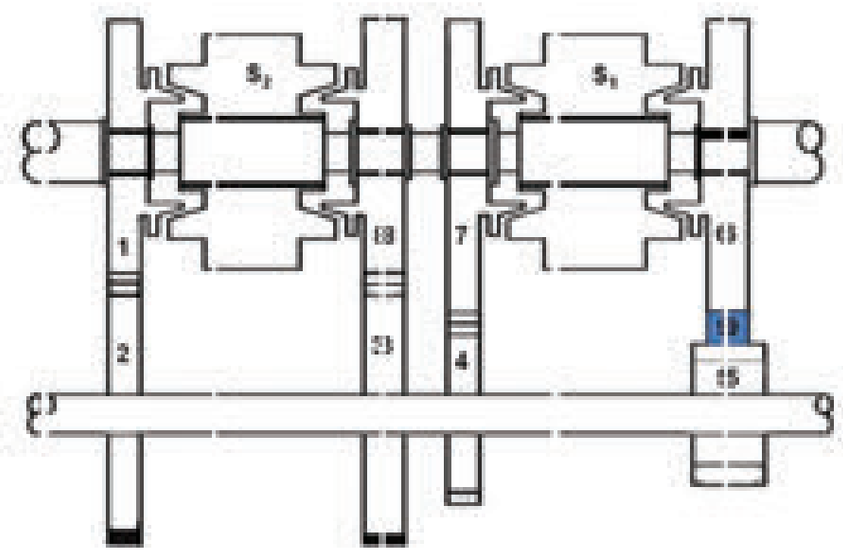


Figure 3.5.3.1 Neutral condition S1 and S2 are held stationary

First gear

By actuating the gear shift lever, the synchronizer S_1 is moved towards left side as shown in Figure 3.5.3.2 to engage gear 7 in the main shaft.

Now the rotational power is transmitted to the main shaft through the dog clutch. Now the speed is low. The power transmission from the clutch shaft to the output shaft is by gear 1-2 is

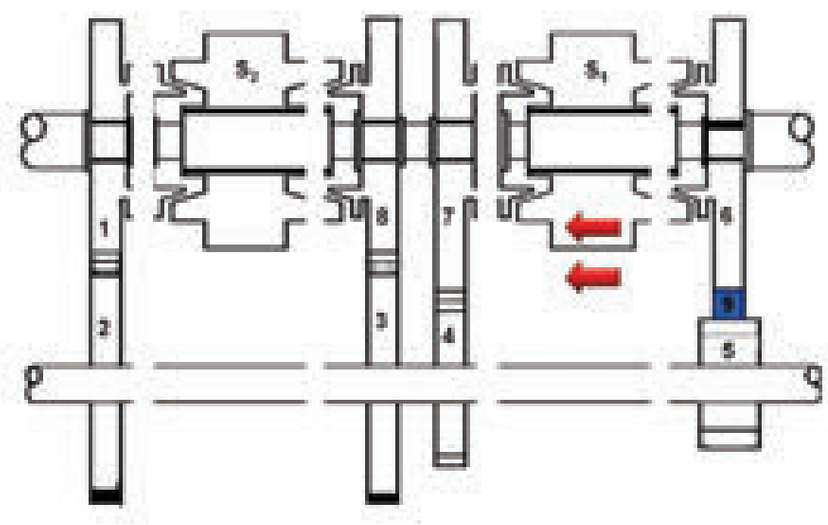


Figure 3.5.3.2 First gear condition

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